Juan Sebastián Pérez

Senior AI Engineer

LLM agents · RAG · Evals · Production AI systems

Bucaramanga, Colombia · Remote (UTC-5)

juans.perezc@gmail.com
linkedin.com/in/juan-perez-ai-engineer
github.com/Juanllenato
juan-perez-ai.vercel.app

7 AI products designed and built

120+ active users on a live AI CRM

1,500+ automated tests across products

2 days to launch a new vertical on the agent core

SUMMARY

Senior AI engineer with 5+ years in software and 3+ years putting LLM systems into production. At AINEATECH I lead AI product engineering and have designed and built seven AI products for companies in Colombia and Peru, including WhatsApp sales agents, a multi-tenant sales SaaS and two AI CRMs (one with 120+ active users). I focus on what makes an agent reliable in production: strict tool schemas, checks in code before risky actions, eval suites and predictable costs.

- Built a reusable agent core (strict-schema tools, output checks, human handoff, multi-tenant Postgres and evals) and reused it in 4 verticals. The car-dealership version launched in **2 days**.
- Architected an AI CRM with **120+ active users** in Colombia and Peru: a RAG assistant over live business data, invoice OCR over WhatsApp and nightly AI reports.
- **1,500+ automated tests** and **75+ eval scenarios** with simulated customers and an LLM judge. The real-estate agent's baseline had **zero data leaks and zero ungrounded figures**.

EXPERIENCE

Senior AI Engineer, Lead of AI Products

2023 – Present

AINEATECH (AINEA Technology LLC) · Houston, TX, USA · Remote

- Own AI product delivery end to end: architecture, prompts and model choice, evals, security, CI/CD and production support.
- Write agent loops on the Claude Messages API with strict Zod tool schemas, versioned prompts and prompt caching. An agent can only confirm an order after the tool call succeeds.
- Check every reply in code before it is sent (data leaks, catalog prices, policy rules), with one retry, a safe fallback and a handoff to a person.
- Route between Claude and open-weight models on Together AI (DeepSeek, Kimi, GLM) by cost and benchmarks, with fallbacks and per-turn cost tracking.
- Isolate client data with Postgres row-level security and AES-256-GCM encryption; integrated WhatsApp Cloud API, Embedded Signup, Conversions API and Marketing API.
- Run production on Docker, Caddy and GitHub Actions with automatic rollback. New agents start as shadow-mode pilots, and every product has ADRs and runbooks.
- Also delivered 4+ web and e-commerce sites and n8n AI automations (Pontebela, PrevenSalud, Grupo Alianza Vital).

Freelance Web Developer

2020 – 2023

Independent · Colombia · Remote

- Built websites and online stores for small businesses on WordPress and WooCommerce, with custom Astra child themes, PHP page templates and mobile-first CSS.
- Ran each project end to end: requirements with the client, design, build, launch and maintenance.

SKILLS

AI & LLM

Agents & tool calling · RAG & hybrid search · Evals · LLM-as-judge · Guardrails · Structured outputs · Prompt caching · Prompt-injection defense · Speech-to-text · OCR / document AI

Models

Anthropic Claude · OpenAI · Together AI · DeepSeek · Kimi · GLM · Voyage embeddings

Backend

TypeScript · Node.js · Fastify · Python · FastAPI · PostgreSQL · Supabase · pgvector · Redis · pg-boss · Celery · Webhooks

Integrations

WhatsApp Cloud API · Meta CAPI & Ads · Shopify · Stripe · Google Workspace · n8n

Frontend & DevOps

React · Next.js · React Native · Docker · GitHub Actions · Vitest · Playwright

AVAILABILITY

Immediate
can start right away

Remote, UTC-5
overlaps US business hours

Contractor or employee
USD contracts welcome

EDUCATION

Systems Engineering
UTS, Colombia · in progress

Technologist in Software Systems Development
UTS, Colombia · completed

LANGUAGES

Spanish
native

English
B1, comfortable with written async work (Slack, PRs, docs)

PORTFOLIO

Selected AI products

Juan Sebastián Pérez
Senior AI Engineer

Live used by real customers · **Deployed** running in production · **Pilot** supervised trial with a real client

Sin Frontera

PILOT

AI sales agent for B2B real estate, Colombia

WhatsApp agent that qualifies leads and books site visits for an industrial real-estate firm (about 100 properties, 1,200–1,500 leads a month). Hybrid search blends SQL filters, Spanish full-text and pgvector embeddings with RRF. 11 agent tools, plus a 12-tool assistant the team uses to query data and approve changes. 43 eval scenarios, 540+ tests, 30 ADRs.

TypeScript · Fastify · React · Supabase · pgvector · Claude · Together AI

sinfrontera.aineatech.com

Ainea

DEPLOYED

WhatsApp AI sales agent SaaS

Multi-tenant SaaS for Latin American small businesses that covers the whole loop: Meta ad, WhatsApp chat, AI sales agent, store order and Conversions API purchase event. Self-serve onboarding with Embedded Signup, Shopify draft orders, Marketing API campaigns and Stripe Billing. Postgres turn queue with per-conversation locks, retries and a dead-letter queue. 760+ tests.

TypeScript · PostgreSQL · WhatsApp Cloud API · Meta CAPI · Shopify · Stripe

app.aineatech.com

Ainea CRM

LIVE

CRM for the sales team

Leads with round-robin routing by product, Google Meet scheduling, email campaigns with an automatic 3-step follow-up, metrics and commissions. The app decides and n8n executes, through an HMAC-signed transactional outbox. Google SSO, 3 roles and a data-deletion flow for Colombia's data protection law (Ley 1581).

n8n · Google Meet · Gmail · Google SSO

app.aineatech.com

Ainea Autos

DEPLOYED

AI platform for used-car dealerships, Colombia & Peru

A 24/7 WhatsApp buyer agent (inventory search, server-side financing calculator, trade-in capture, test-drive booking), a web CRM, and a 27-tool assistant for owners and salespeople that proposes actions for a person to confirm and sends proactive alerts. Rule-based lead scoring; inventory synced from Google Sheets.

TypeScript · Supabase · Claude · Google Sheets

autos.aineatech.com

BACCA restaurant agent

PILOT

3 locations, Bucaramanga, Colombia

WhatsApp agent on Claude for delivery, pickup and table bookings. 13 strict tools and 4 deterministic guards: every price must match the menu or a computed total. Bookings respect seat capacity with Postgres advisory locks and slot holds. Understands voice notes and reads menus from photos and PDFs. New restaurants are added as config. 240+ tests, 13 eval scenarios.

Claude · Deepgram · PostgreSQL · WhatsApp Cloud API

PrevenSalud AI CRM

LIVE · 120+ USERS

Multi-tenant AI CRM, Peru

AI CRM in production with 120+ active users. A RAG assistant answers questions over live business data without sending personal data to the model. Invoices sent over WhatsApp go through a cost-first OCR pipeline (Tesseract, then Vision, then Claude), and AI-written PDF reports go out every night.

Python · FastAPI · PostgreSQL RLS · Redis · Celery · Next.js · Docker

crm.prevensalud.pe

Agentic AI health coach

GITHUB

Mobile app, code on GitHub

An AI coach that runs a whole React Native app through chat, orchestrating 13 tools: RAG on pgvector, meal logging from photos and health projections. Modular FastAPI backend with per-tenant feature flags.

React Native · FastAPI · pgvector

github.com/Juanllenato/plataforma-bienestar-ai

LLM eval harness & n8n automations

GITHUB

Open source

An eval harness with versioned datasets, LLM-as-judge and CI gating. n8n workflows for support triage, BANT lead scoring and invoice approval with fraud scoring and a human in the loop.

LLM-as-judge · CI gating · n8n

github.com/Juanllenato

Case studies, architecture diagrams and an AI assistant that answers questions about my work — **juan-perez-ai.vercel.app**